

# Jin Hyuk Kim

M.S. Student in Computer Engineering

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## Research Interests

Interests    📌 AI4Science for Drug Discovery, Molecular Generation, Agentic AI, Molecular Representation Learning, Explainable/Interpretable AI

## Education

- 2025.03 – Present    📌 **Hallym University**  
*M.S. in Computer Engineering*  
[AI-Powered Cheminformatics Laboratory \(APC Lab\)](#) (Advisor: [Jonghwan Choi](#))
- 2019.03 – 2025.02    📌 **Hallym University**  
*B.S. in Big Data, Division of Software / Double Major in Smart IoT*  
Capstone Project: [CanChem](#)

## Publications

\* Corresponding author

### International Journal

- 1 **Jin Hyuk Kim**, Gyeong Hwan Kim, Hyeon Jun Park, Jonghwan Choi\*, “CUE: A Chemical Uncertainty-Aware Embedding Framework for Multi-modal Drug Selectivity Prediction,” 2026. (**On Revision**)
- 2 **Jin Hyuk Kim**, Jonghwan Choi\*, “OCSAug: Diffusion-based Optical Chemical Structure Data Augmentation for Improved Hand-drawn Chemical Structure Image Recognition,” *The Journal of Supercomputing*, 2025.

### International Conference

- 1 **Jin Hyuk Kim**, Jonghwan Choi\*, “Agentic Multi-objective Molecular Optimization via Dynamic Routing of Property-specific Editor Networks,” in *AI4Sci Conference Korea*, 2026. (**Submitted**)

### Domestic Journal

- 1 Tae Woong Song, **Jin Hyuk Kim**, Hyun Jun Park, Jonghwan Choi\*, “A Study on Semantic Molecular Prompt Engineering Method for Large Language Model-based Molecular Property Prediction,” *KIISE Transactions on Computing Practices*, 2026.
- 2 Tae Woong Song, **Jin Hyuk Kim**, Hyun Jun Park, Jonghwan Choi\*, “Pretrained Large Language Model-based Drug-Target Binding Affinity Prediction for Mutated Proteins,” *Journal of KIISE*, 2025.

### Domestic Conference

- 1 Hyun Jun Park, **Jin Hyuk Kim**, Tae Woong Song, Jonghwan Choi\*, “M-WPDTA: Mutation-Aware Drug-Target Affinity Prediction with Wildtype Referenced Dual-Encoder Modeling,” in *Korea Computer Congress (KCC)*, Poster, 2025.
- 2 Tae Woong Song, **Jin Hyuk Kim**, Hyun Jun Park, Jonghwan Choi\*, “A Study on Molecular Prompt Engineering Method for Large Language Model-based Molecular Property Prediction,” in *Korea Computer Congress (KCC)*, Poster, Outstanding Presentation Paper Award, 2025.
- 3 **Jin Hyuk Kim**, Tae Woong Song, Jonghwan Choi\*, “A Study on DDPM-based Molecular Generation and Semi-Supervised Learning for Improving the Performance of Optical Chemical Structure Recognition,” in *Annual Symposium of KIPS (ASK)*, Oral, 2024.

## Teaching Experience

- Jul. 2025

Teaching Assistant, AI-Driven Drug Discovery Program for High School Students

Hallym University, supported by the Gangwon State Office of Education
- Spring 2025

Teaching Assistant, Reinforcement Learning

Hallym University
- Fall 2024

Teaching Assistant, Database Basics

Hallym University

## Additional Information

### Patents

- 2025.04.28

키나아제에 대한 화합물의 선택성을 예측하는 전자 장치의 동작 방법

최종환, 김진혁

대한민국 특허출원 10-2025-0055457
- 사전학습된 거대 언어 모델을 이용하여 돌연변이 단백질에 대한 약물의 결합 친화도를 예측하는 전자 장치, 방법 및 프로그램

최종환, 송태웅, 김진혁, 박현준

대한민국 특허출원 10-2025-0055458

### Honors & Awards

- 2026.05

Outstanding Graduate Research Paper Award

Presented at the 44th Anniversary Ceremony of Hallym University

### Certifications

- 2026.06

ADsP: Advanced Data Analytics Semi-Professional

Korea Data Agency